

AUBURN, CA, USA

WMO: 720267

Lat: 38.955N

Lon: 121.082W

Elev: 467

StdP: 95.84

Time Zone: -8.00 (NAP)

Period: 04-19

WBAN: 23224

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.0	1.2	-9.0	1.9	9.3	-7.2	2.2	8.3	8.8	8.6	7.9	9.2	1.1	100	0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	37.1	19.4	35.0	18.4	33.6	18.1	20.7	33.4	19.9	32.1	19.1	31.0	2.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.4	12.3	24.9	15.2	11.4	24.7	14.0	10.6	23.0	61.7	33.1	58.5	32.3	55.9	31.0	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	5.6	4.9	-2.7	39.9	2.0	3.6	-4.1	42.5	-5.2	44.6	-6.3	46.5	-7.8	49.1		
(5)			-3.7	22.4	1.6	1.9	-4.9	23.8	-5.9	24.9	-6.8	25.9	-8.0	27.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	8.5	9.4	11.3	13.2	17.3	22.1	26.0	25.4	22.7	17.1	11.8	7.8	(6)
(7)		DBStd	7.33	3.03	3.46	3.61	4.02	4.11	4.34	3.15	3.00	4.10	3.83	3.57	2.82	(7)
(8)		HDD10.0	253	65	48	27	12	1	0	0	0	0	1	21	79	(8)
(9)		HDD18.3	1620	304	249	218	162	69	14	0	0	11	70	196	326	(9)
(10)		CDD10.0	2479	19	32	68	108	227	364	496	476	382	220	76	11	(10)
(11)		CDD18.3	804	0	0	1	8	37	128	237	218	142	31	1	0	(11)
(12)		CDH23.3	9358	1	4	9	77	417	1579	2937	2537	1503	282	11	0	(12)
(13)		CDH26.7	4499	0	1	1	11	126	732	1563	1288	708	71	1	0	(13)
(14)	Wind	WSAvg	2.2	1.9	2.4	2.5	2.6	2.5	2.3	2.4	2.3	2.2	1.9	1.8	1.9	(14)
(15)	Precipitation	PrecAvg	900	157	132	145	69	18	7	1	4	19	59	150	138	(15)
(16)		PrecMax	1648	431	447	388	196	72	28	23	40	94	352	342	360	(16)
(17)		PrecMin	299	13	7	29	2	0	0	0	0	0	0	18	0	(17)
(18)		PrecStd	301	109	104	92	59	19	8	4	9	27	70	97	93	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	22.2	24.1	27.9	32.6	37.7	38.9	37.8	37.1	32.1	25.0	17.9	(19)
(20)			MCWB	10.7	13.4	14.8	17.1	18.9	20.2	20.9	19.0	18.4	16.4	14.4	9.9	(20)
(21)		2%	DB	17.8	19.2	22.3	26.0	29.1	35.2	37.1	36.2	34.0	28.0	22.6	16.2	(21)
(22)			MCWB	10.0	12.0	13.7	16.0	17.2	19.2	19.7	18.3	17.1	15.1	13.4	9.8	(22)
(23)		5%	DB	16.2	17.5	20.2	22.9	27.6	32.8	35.1	34.0	32.6	26.4	21.0	13.9	(23)
(24)			MCWB	9.7	10.7	12.7	14.2	16.2	18.1	18.7	18.1	16.9	14.1	12.5	9.3	(24)
(25)		10%	DB	13.7	15.9	17.8	21.0	26.0	31.2	33.0	32.6	31.2	24.0	17.7	12.6	(25)
(26)			MCWB	9.1	10.0	11.4	13.6	15.5	17.5	18.4	17.6	16.6	13.5	11.3	9.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	14.4	15.7	18.5	19.9	22.1	22.6	20.4	19.5	17.9	16.9	13.7	(27)
(28)			MCDWB	15.5	17.6	22.7	26.3	29.9	34.7	36.6	34.1	30.1	25.9	22.4	14.6	(28)
(29)		2%	WB	12.1	13.0	14.3	16.7	18.5	20.7	20.8	19.7	18.4	16.7	14.9	12.4	(29)
(30)			MCDWB	14.8	17.7	20.5	24.2	28.5	32.1	34.3	32.9	31.2	26.0	19.8	13.2	(30)
(31)		5%	WB	11.0	11.9	13.2	15.2	17.2	19.5	20.1	18.9	17.6	15.6	13.5	11.3	(31)
(32)			MCDWB	13.5	15.4	18.4	21.7	25.5	30.6	33.0	32.1	30.5	23.6	18.0	12.4	(32)
(33)		10%	WB	10.1	11.0	12.1	13.9	16.1	18.4	19.2	18.3	16.9	14.6	12.4	10.1	(33)
(34)			MCDWB	13.2	14.1	16.6	19.7	24.0	28.5	31.5	31.0	29.3	21.7	16.2	12.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.6	8.9	9.2	10.2	11.7	13.4	13.6	13.5	12.9	11.3	9.5	8.4	(35)
(36)			MCDBR	12.2	12.4	12.1	12.6	13.6	14.9	14.3	13.9	13.8	13.4	12.7	10.6	(36)
(37)			MCWBR	7.3	7.8	7.4	7.5	6.7	6.1	5.6	5.8	6.2	6.3	6.8	6.5	(37)
(38)		5% WB	MCDWB	9.7	10.3	10.7	11.6	12.8	13.8	13.4	13.3	13.2	11.4	10.7	6.8	(38)
(39)			MCWBR	6.6	7.0	6.9	7.2	6.5	6.0	5.5	5.7	6.0	5.8	6.5	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.310	0.338	0.342	0.357	0.349	0.348	0.332	0.341	0.332	0.310	0.323	(40)	
(41)		taud	2.514	2.525	2.433	2.424	2.404	2.441	2.443	2.478	2.459	2.483	2.525	2.450	(41)	
(42)		Ebn at Noon	871	921	922	934	921	925	923	933	905	880	861	818	(42)	
(43)		Edh at Noon	77	86	105	112	117	113	112	105	101	89	75	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.21	3.14	4.47	5.93	7.25	8.27	8.25	7.40	5.96	4.18	2.66	1.93	(44)	
(45)		RadStd	0.37	0.43	0.53	0.57	0.52	0.38	0.29	0.26	0.23	0.30	0.21	0.33	(45)	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.40	N/A	-1.09	N/A	N/A	N/A	-114	+138

Nomenclature: See separate page